

ORIGINAL ARTICLE

Association between CT-detected brain metastases and EGFR mutation status in non-small cell lung cancer

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ABSTRACT

Background and aim: To evaluate the relationship between head multidetector computed tomography (MDCT) features of brain metastases and epidermal growth factor receptor (EGFR) mutation status in patients with histologically confirmed non-small cell lung carcinoma (NSCLC).

Methods: In this retrospective cross-sectional study, medical records and PACS imaging from Dr. Wahidin Sudirohusodo General Hospital (January 2022–May 2025) were screened. Of 1,096 patients with lung tumors, 33 met inclusion criteria (histopathologically confirmed NSCLC, available EGFR result, and evaluable head MDCT demonstrating brain metastasis). Head MDCT scans were reviewed for number of lesions, rim enhancement, peritumoral vasogenic edema, hemorrhagic component, leptomeningeal disease and calcification.

Results: Patients were classified as EGFR-mutant ($n = 14$) or EGFR-wild-type ($n = 19$). The cohort was 60.6% male, most patients were aged 56–65 years, and 51.5% were never-smokers. Rim enhancement on MDCT differed by EGFR status: EGFR-wild-type lesions more often demonstrated rim enhancement (94.7% of wild-type vs. 42.9% of mutant lesions), while EGFR-mutant lesions more often lacked rim enhancement (57.1%); this association was statistically significant ($p = 0.002$). No significant relationships were found between EGFR status and number of lesions, vasogenic edema, hemorrhagic change, leptomeningeal involvement or calcification



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(all $p > 0.05$). NSCLC histologic subtype and EGFR mutation subtype analyses similarly showed no statistically significant associations with the assessed MDCT features.

Conclusions: This study showed rim enhancement on head MDCT was associated with EGFR status, being more frequent in EGFR-wild-type tumors; other MDCT features were not significantly related to EGFR mutation. Larger, prospective studies using MRI and comprehensive molecular profiling are warranted to validate these observations. (www.actabiomedica.it)

Key words: brain metastasis, non-small cell lung carcinoma, EGFR, head CT scan

Introduction

Lung cancer is the leading cause of cancer mortality and the second most frequently diagnosed malignancy worldwide, with an estimated 2.2 million new cases and 1.8 million deaths in 2020. Prognosis remains poor: five-year overall survival is under 20% in most countries. In 2018 the global lung cancer mortality was 1,761,000, of which China accounted for 690,000 deaths (1,2). The burden in Indonesia is substantial. Lung cancer was the leading cause of cancer death in 2018, responsible for 12.6% of all cancer deaths and 8.6% of cancer cases that year. It ranks among the most common cancers nationally, behind breast, cervical and colorectal cancers, and the annual case load is projected to rise from 30,023 in 2018 to 54,983 by 2040. Notably, approximately 80% of smokers aged ≥ 15 years live in low- and middle-income countries, exacerbating future cancer risk (1,2). Brain metastases are the most common intracranial neoplasm in adults. Population incidence is reported at about 14 per 100,000 per year, substantially higher than that of primary brain tumors; among cancer patients overall, brain metastasis rates are estimated at 8.5–9.6%, most often due to hematogenous spread. The commonest primary sources are lung (39–56%), breast (13–30%), skin (8–11%), gastrointestinal (6–9%), and kidney (2–6%) cancers, with 2–14% of brain metastases presenting without an identifiable primary tumor (3). Brain metastasis is a frequent and prognostically adverse complication of lung cancer. In non-small cell lung cancer (NSCLC),

reported incidence ranges from 20% to 40%. Improvements in systemic therapies have extended survival, which paradoxically increases the cumulative risk of developing brain metastases. Contemporary data on brain metastasis incidence are incomplete, particularly in the era of molecularly targeted therapies, and tumor biology differences likely drive heterogeneity in metastatic patterns and individual risk (4). Mutations in the epidermal growth factor receptor (EGFR) tyrosine kinase domain detected in roughly 15% of patients with advanced non-squamous NSCLC are routinely assayed in clinical practice and serve as targets for tyrosine kinase inhibitors (TKIs) used in patients with brain metastases. Several studies report a higher propensity for brain metastasis among EGFR-mutant tumors compared with EGFR wild-type tumors, whereas other investigations have found no significant difference between EGFR statuses (4,5). Computed tomography (CT) is commonly employed to detect and characterize intracranial metastatic lesions, providing information on number, location and radiographic features. However, the relationship between CT-characterized brain metastasis features and EGFR mutation status in NSCLC is not fully defined. At Wahidin Sudirohusodo General Hospital, EGFR testing is not performed on-site and specimens must be sent to an external laboratory. Our null hypothesis that there is no association between brain metastasis and epidermal growth factor receptor (EGFR) status in patients with non-small cell lung carcinoma (NSCLC). The alternative hypothesis posits that brain metastasis is associated with EGFR

status in NSCLC, with a higher prevalence observed in patients harboring EGFR mutations compared with those with wild-type EGFR, and that the characteristics of brain metastases differ according to EGFR status. This study therefore aims to describe EGFR mutation status among NSCLC patients with brain metastases at our centre and to evaluate the association between EGFR status and CT characteristics of intracranial metastatic disease.

Materials and methods

Study design and patient characteristics

This retrospective cross-sectional observational study used secondary data to evaluate the relationship between CT-detected brain metastasis characteristics and EGFR mutation status in patients with NSCLC. The study was conducted at Dr. Wahidin Sudirohusodo General Hospital, Makassar. Imaging data were retrieved from the hospital Picture Archiving and Communication System (PACS) and clinical information from medical records for patients imaged between January 2022 and May 2025. All patients with available and eligible records during the study period who met the inclusion criteria were included by consecutive sampling. Eligible cases were identified through a search of the PACS and the hospital medical record system; imaging and clinical data were abstracted retrospectively. Inclusion criteria were histopathological diagnosis of NSCLC confirmed by anatomical pathology; at least one multidetector computed tomography (MDCT) scan of the head performed with or without contrast; and documented EGFR mutation testing results. Exclusion criteria were incomplete clinical or imaging records that precluded evaluation and MDCT examinations showing other predominant intracranial pathology (for example acute trauma or intracranial infection) that would confound assessment of metastatic lesions.

Research protocol

Demographic and clinical variables extracted from the medical record included age, sex, smoking

history, prior therapies, and tumor stage. Head MDCT examinations were reviewed in the PACS environment. Imaging review was performed by experienced radiologists (blinded to EGFR status), reporting the number, size, location, contrast enhancement pattern and other relevant radiographic features of intracranial metastatic lesions. Discrepant readings were resolved by consensus. EGFR mutation status was ascertained from pathology reports in the medical record. When EGFR testing was performed off-site, test reports returned to the patient record were used for classification. Patients were grouped as EGFR-mutant or EGFR-wild-type according to the documented molecular result. The study protocol was reviewed and approved by the Biomedical Health Research Ethics Committee, Faculty of Medicine, Hasanuddin University, Makassar (No. 132/UN4.5.4.5.31/PP36/2025). Because this investigation used de-identified retrospective data, the requirement for individual informed consent was waived by the ethics committee.

Statistical analysis

Univariate analyses described baseline characteristics and imaging findings. Categorical variables are presented as counts and percentages. Normality of continuous variables was assessed using the Shapiro–Wilk test. Bivariate analysis was utilized to assess the possible correlation between brain metastasis characteristics and epidermal growth factor receptor (EGFR) status in patients diagnosed with non-small cell lung carcinoma (NSCLC). Chi-square test was applied when the data satisfied its underlying assumptions, including adequate expected cell counts, with no more than 20% of cells having an expected value of less than five. In instances where these assumptions were not met, alternative statistical approaches were employed. Fisher's exact test was applied to 2×2 contingency tables, while the Kolmogorov–Smirnov test was used for $2 \times K$ tables. Statistical significance was defined as a P-value of less than 0.05. Analyses were performed using standard statistical software SPSS v25.0. Results were presented narratively and supplemented with tables and figures where appropriate.

Results

Of 1,096 patients with a diagnosis of lung tumor identified in the hospital database, 203 had undergone head MDCT. After review of histopathology and imaging availability, and application of the study inclusion and exclusion criteria, 33 patients with histologically confirmed non-small cell lung carcinoma (NSCLC), available EGFR results and evaluable head MDCT scans were included in the final analysis (see Table 1 for sample characteristics). Cases were excluded at earlier steps due to lack of head MDCT, non-NSCLC histopathology, absence of intracranial metastatic disease on MDCT, unavailable EGFR testing, or non-evaluable imaging (Figure 1).

The demographic and clinical features of the 33 included patients are summarized in Table 1. Males comprised 60.6% of the sample ($n = 20$). The largest age group was 56–65 years (48.5%, $n = 16$). A slight majority had no history of smoking (51.5%, $n = 17$). The most frequent clinical stage at presentation was IVA (51.5%, $n = 17$).

Imaging findings stratified by EGFR status (mutation versus wild-type) are presented in Table 2. Multiple intracranial metastatic lesions were the predominant pattern in both EGFR-mutation and EGFR-wild-type groups (57.1% and 63.2%, respectively); there was no statistically significant association between EGFR status and the number of brain metastases ($p = 0.727$). Rim enhancement (Figure 2) on MDCT differed by EGFR status: most EGFR-mutant cases lacked rim enhancement (57.1%, $n = 8$), whereas rim enhancement was observed in the majority of EGFR-wild-type cases (94.7%, $n = 18$). This association was statistically significant ($p = 0.002$) (Figure 3). Vasogenic edema was frequent across groups (100% of EGFR-mutation cases and 89.5% of EGFR-wild-type cases) and was not significantly associated with EGFR status ($p = 0.324$). The presence of hemorrhagic components did not differ significantly between EGFR groups (absence of hemorrhage in 85.7% of mutant and 89.5% of wild-type cases; $p = 0.574$). No cases in either EGFR group demonstrated leptomeningeal metastasis (Figure 4). Calcification

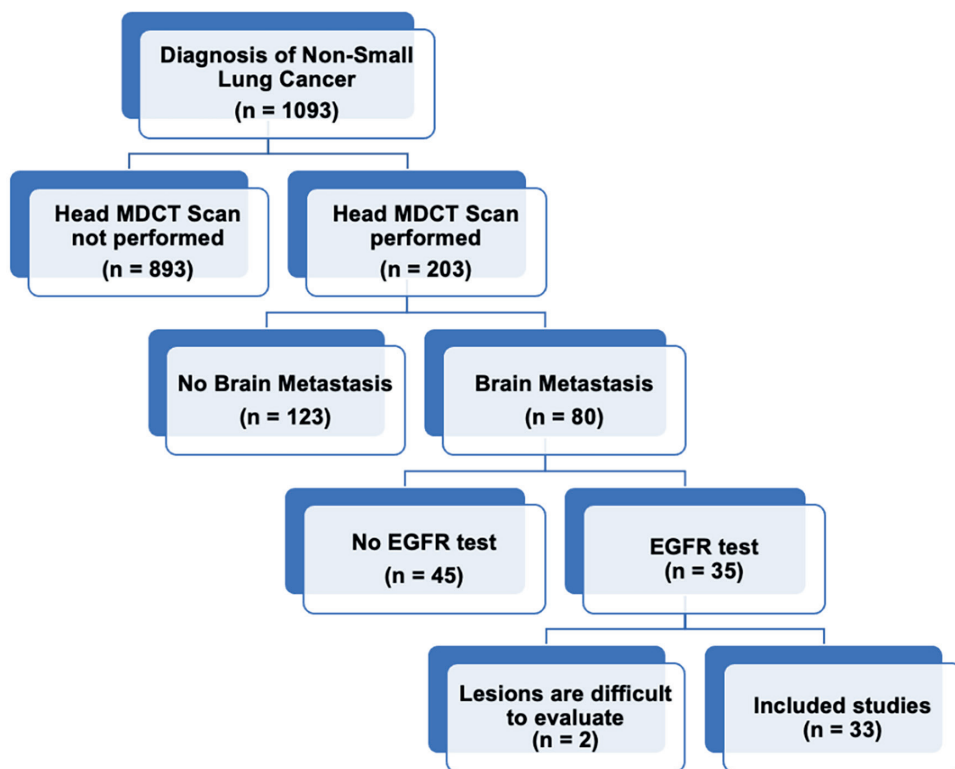


Figure 1. Patient selection flowchart.

(Figure 5) was uncommon and similarly distributed between groups (78.6% of mutants and 78.9% of wild-type cases showed no calcification; $p = 0.652$).

This study summarizes the distribution of NSCLC histologic subtypes and their relationship

to brain metastasis in the broader cohort and in the analyzed sample. In the examined cohort, histologic subtype (adenocarcinoma, squamous cell carcinoma, adenosquamous carcinoma) was not significantly associated with the presence of brain metastasis ($p = 0.589$). (Table S1)

The distribution of imaging features by EGFR mutation subtype (Exon 19 versus Exon 21) is shown in Table 3. Exon 19 mutations were more frequently observed in solitary lesions (54.5%, $n = 6$), while Exon 21 mutations were observed exclusively in cases with multiple lesions in this sample (100%). These differences were not statistically significant ($p = 0.154$). Regarding rim enhancement, Exon 19 cases were more often without rim enhancement (72.7%, $n = 8$), whereas all Exon 21 cases showed rim enhancement (100%); this comparison approached but did not reach conventional significance ($p = 0.055$). All cases in both Exon 19 and Exon 21 groups exhibited vasogenic edema. Hemorrhagic components, leptomeningeal disease and calcification did not differ significantly by mutation subtype ($p = 0.604$, no leptomeningeal cases, and $p = 0.453$, respectively).

Relationship between NSCLC histologic type and MDCT features are presented in Table 4. Multiple metastases were most common in adenocarcinoma

Table 1. Baseline patient characteristics.

Variables	n (%)
Gender	
Male	20 (60.6)
Female	13 (39.4)
Age (years)	
26-35	2 (6.1)
36-45	4 (12.1)
46-55	8 (24.2)
56-65	16 (48.5)
>65	3 (9.1)
Smoking history	
Yes	16 (48.5)
No	17 (51.5)
Stage	
IVA	17 (51.5)
IVB	16 (48.5)

Table 2. Relationship between characteristics of brain metastatic lesions and EGFR status.

Metastatic lesion characteristics		EGFR, n (%)		p value
		Mutation type (n = 14)	Wild type (n = 19)	
No. of lesions	Solitary	6 (42.9)	7 (36.8)	0.727
	Multiple	8 (57.1)	12 (63.2)	
Rim enhanced	Yes	6 (42.9)	18 (94.7)	0.002
	No	8 (57.1)	1 (5.3)	
Vasogenic Edema	Yes	14 (100)	17 (89.5)	0.324
	No	0 (0)	2 (10.5)	
Hemorrhagic Lesion	Yes	2 (14.3)	2 (10.5)	0.574
	No	12 (85.7)	17 (89.5)	
Leptomeningeal Metastasis	Yes	0 (0)	0 (0)	N.A
	No	14 (100)	19 (100)	
Calcification	Yes	3 (21.4)	4 (21.1)	0.652
	No	11 (78.6)	15 (78.9)	

*Values are n (%). Categorical variables were compared with Chi-square test.

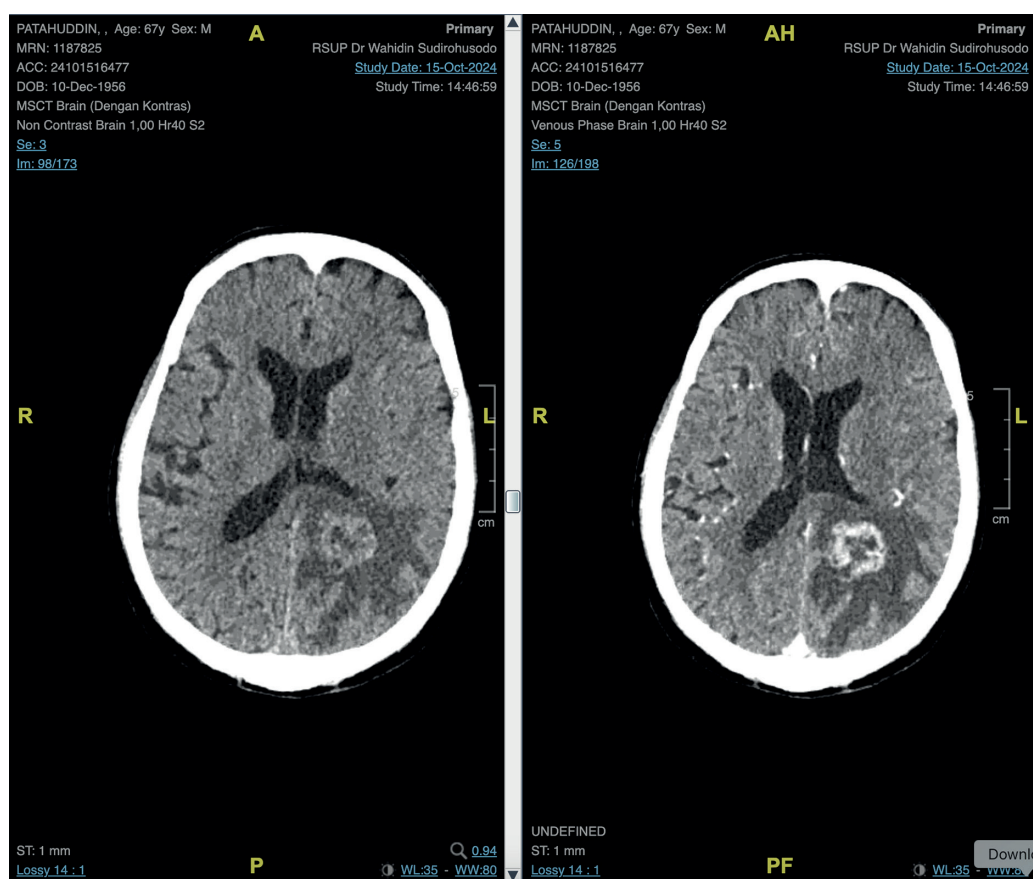


Figure 2. A 69-year-old man was diagnosed with squamous cell carcinoma lung tumor. An MDCT brain scan was performed with the following results: a) heterogeneous lesion (25-35 HU) (red arrow) which in b) post-contrast shows rim enhancement (72 HU) (dashed red arrow) in the left parietal lobe with clear boundaries, irregular edges, non-calcification that presses and narrows the left lateral ventricle medially accompanied by extensive vasogenic edema around it. No hemorrhagic lesions or leptomeningeal enhancement were seen. Pathology results showed EGFR non-mutation (wild type).

Table S1. Relationship between the number of metastatic brain lesion characteristics and NSCLC types

Variables		NSCLC, n (%)		p value
		Adenocarcinoma (n = 69)	SCC (n = 93)	
Metastatic lesion	Yes	32 (46.3)	38 (40.86)	0.589
	No	37 (56.32)	55 (59.13)	

*Values are n (%). Categorical variables were compared with Chi-square test.

(68.2%, n = 15), whereas solitary lesions predominated in squamous cell carcinoma (71.4%, n = 5); adenosquamous carcinomas most often had multiple lesions (75%, n = 3). No statistically significant associations were observed between histologic type

and number of lesions (p = 0.872), rim enhancement (p = 0.998), vasogenic edema (p = 0.972), hemorrhagic change (p = 1.000), or calcification (p = 1.000). Leptomeningeal metastasis was not observed in any histologic subgroup.

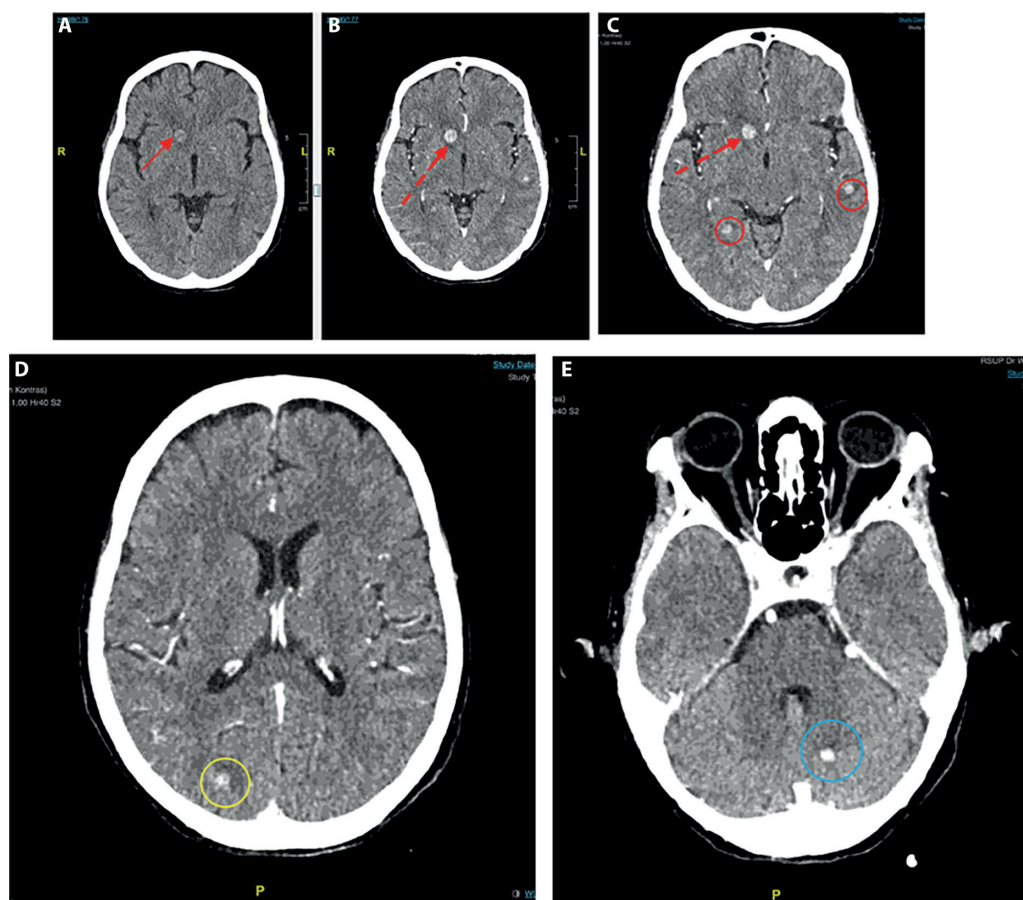


Figure 3. A 52-year-old man diagnosed with adenocarcinoma lung tumor. An MDCT brain scan was performed with the following results: a) Multiple isodense lesions (38 HU) (red arrow) and b,c) which, with post-contrast, appear as homogeneous enhancement (74 HU) in a relatively round shape, with clear boundaries, irregular edges, and non-calcification in the right caudate nucleus (dashed red arrow), left temporal lobe (red circle), d) right occipital lobe (yellow circle), e) left cerebellum (blue circle), accompanied by minimal vasogenic edema around them. No hemorrhagic lesions or leptomeningeal enhancement were seen. Pathology results showed EGFR mutation type (Exon 19).

Table 5 displays EGFR status distribution across NSCLC histologic types: adenocarcinomas were more frequently EGFR-wild-type in this sample (68.2%, $n = 15$), whereas squamous cell and adenosquamous carcinomas were more often EGFR-mutant (57.1% and 75%, respectively). These differences were not statistically significant ($p = 0.509$). Table 6 details EGFR mutation subtypes by histology: Exon 19 predominated among adenocarcinomas (85.7% of mutated adenocarcinomas), squamous cell carcinomas showed both Exon 19 and Exon 21 mutations equally (50% each among mutated SCC cases), and all mutated adenosquamous cases harbored Exon 19 (100%).

There was no significant association between mutation subtype and histologic type ($p = 0.995$).

Discussion

In this retrospective single-center series of 33 patients with histologically confirmed NSCLC and evaluable head MDCT, the only imaging characteristic that demonstrated a statistically significant association with EGFR status was rim enhancement, which was predominantly observed in EGFR-wild-type tumors, whereas EGFR-mutant tumors more often

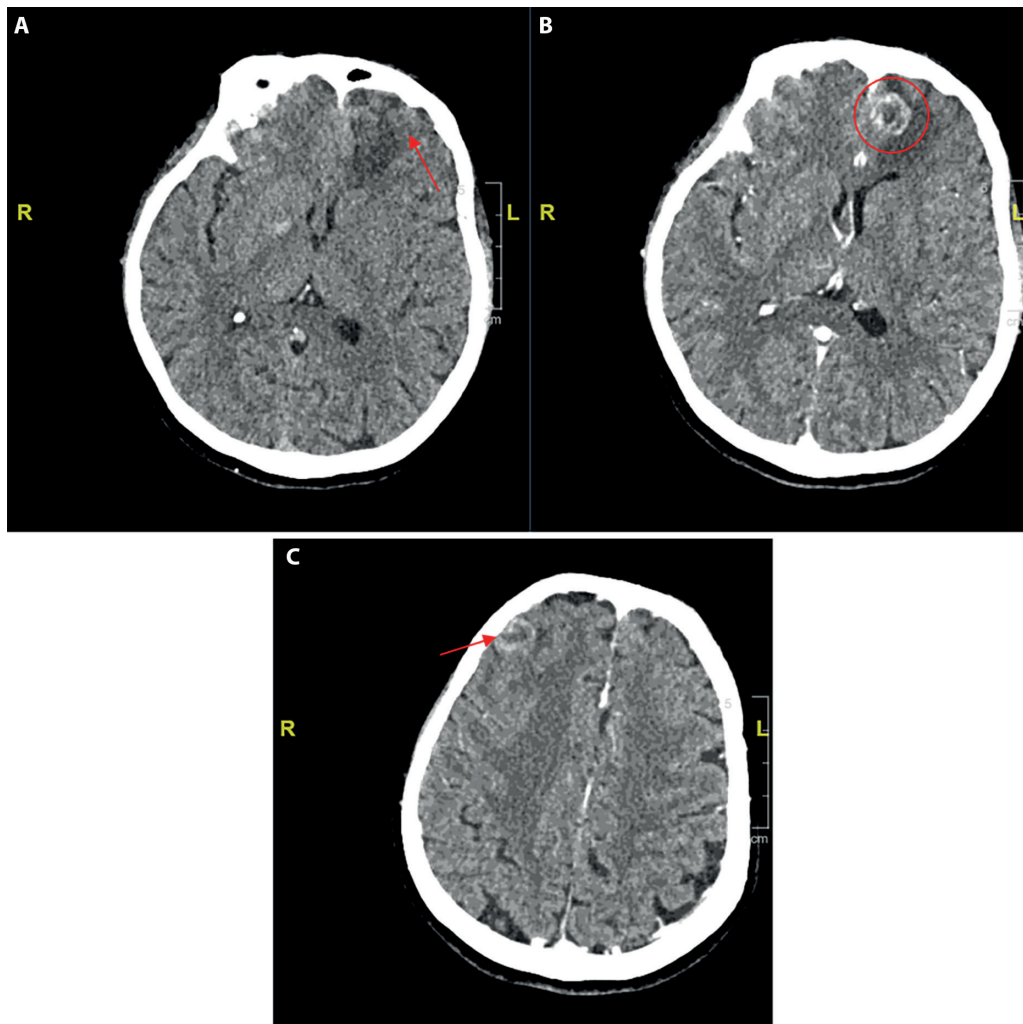


Figure 4. A 46-year-old woman diagnosed with adenocarcinoma lung tumor. An MDCT brain scan was performed with the following results: a) Multiple heterogeneous lesions (18-41 HU) which, after contrast administration, b, c, d) showed rim enhancement (74 HU), clear boundaries, irregular edges, relatively round in shape, in the right frontal lobe (red arrow), left frontal lobe (red circle) accompanied by vasogenic edema around them. No hemorrhagic lesions or leptomeningeal enhancement were observed. Pathology results showed an EGFR mutation type (Exon 21).

lacked rim enhancement. This finding aligns partially with prior observations that tumor genotype may influence radiographic appearance, although the literature is heterogeneous. Several studies have reported no consistent differences in the overall burden or distribution of brain metastases between EGFR-mutant and EGFR-wild-type NSCLC, while others have identified genotype-specific patterns for particular mutation subtypes (6–9). For example, Sekine et al.,(7) reported that EGFR exon 19 deletions were

associated with multiple, smaller metastases and less peritumoral edema compared with wild-type tumors, whereas L858R (Exon 21) cases did not differ significantly from wild-type. In the present study, comparisons by mutation subtype (Exon 19 versus Exon 21) did not reach statistical significance, although trends in rim enhancement by exon (absence of rim enhancement in Exon 19 and presence in Exon 21) were observed and approached significance. Demographically, our sample had a predominance of males

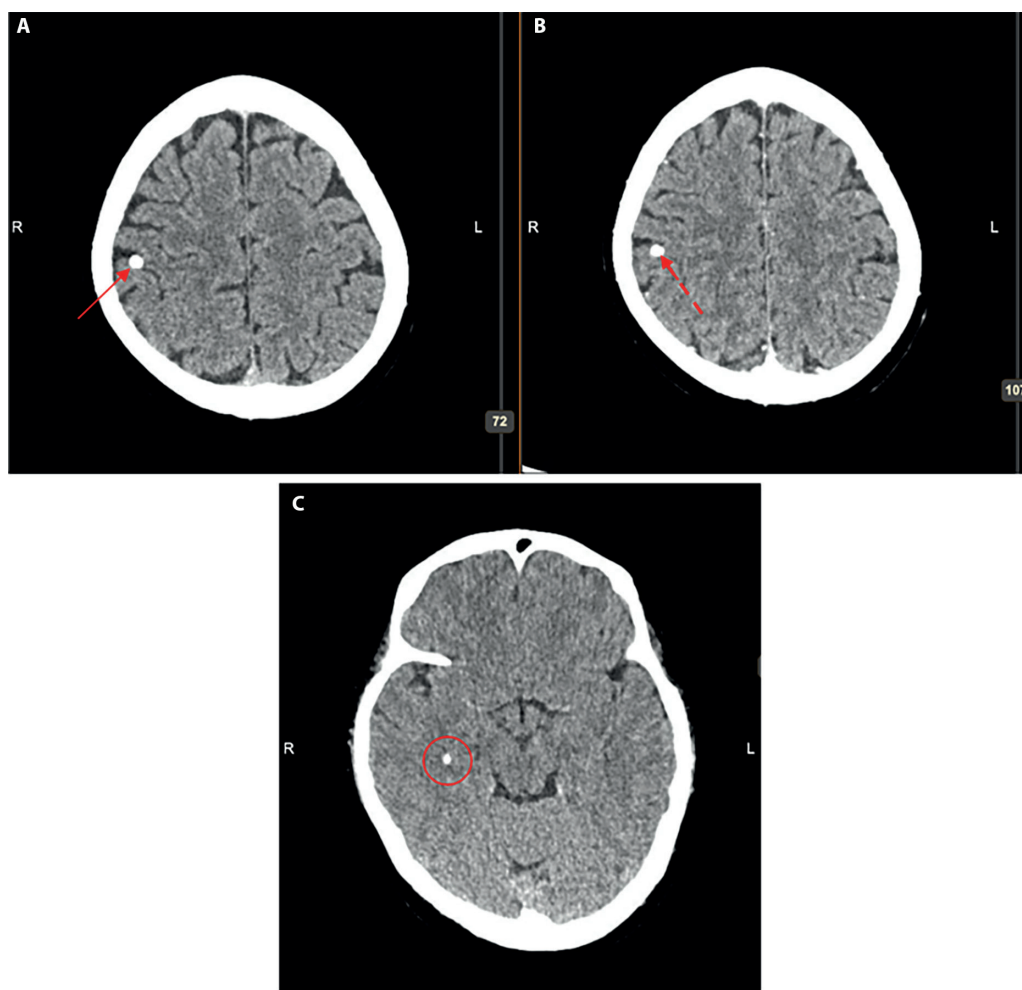


Figure 5. A 65-year-old woman diagnosed with adenocarcinoma lung tumor. MDCT brain scan results: a) Multiple calcifications (626 HU) (red arrow) that did not enhance post-contrast b,c) in the right frontal lobe (dashed red arrow), right temporal lobe (red circle). No rim enhancement, hemorrhagic lesions, vasogenic edema, or leptomeningeal enhancement were observed. The PA results showed EGFR non-mutation (wild type).

(60.6%) and an age distribution centered in the sixth decade, consistent with other series of NSCLC with brain metastases (10). The observed higher proportion of non-smokers in the present sample is notable and consistent with reports that EGFR-mutant NSCLC is more frequent among non-smokers and among women, who have been shown to derive particular benefit from EGFR-targeted therapies (10,11). Stage distribution in our cohort (majority stage IVA) mirrors the expected association between advanced primary disease and intracranial dissemination (12). The absence of significant associations between EGFR status

and other CT features assessed in this study including number of lesions, vasogenic edema, hemorrhagic change, leptomeningeal involvement and calcification is concordant with multiple prior reports that found heterogeneous or null relationships between genotype and these imaging characteristics (6,8). Differences in study design, imaging modality (CT versus MRI), timing of imaging relative to systemic therapy, and small subgroup sizes likely contribute to inconsistent findings across studies. For example, hemorrhagic propensity has been variably linked to histology and lesion biology, with some studies suggesting higher

Table 3. Relationship between the characteristics of metastatic lesions to the brain and EGFR mutation type.

Metastatic lesion characteristics		EGFR mutation type, n (%)		p value
		Exon 19 (n = 11)	Exon 21 (n = 3)	
No. of lesions	Soliter	6 (54.5)	0 (0)	0.154
	Multiple	5 (45.5)	3 (100)	
Rim enhanced	Yes	3 (27.3)	3 (100)	0.055
	No	8 (72.7)	0 (0)	
Vasogenic Edema	Yes	11 (100)	3 (100)	N.A
	No	0 (0)	0 (0)	
Hemorrhagic Lesion	Yes	2 (18.2)	0 (0)	0.604
	No	9 (81.8)	3 (100)	
Leptomeningeal Metastasis	Yes	0 (0)	0 (0)	N.A
	No	11 (100)	3 (100)	
Calcification	Yes	3 (27.3)	0 (0)	0.453
	No	8 (72.7)	3 (100)	

*Values are n (%). Categorical variables were compared with Fisher exact test.

Table 4. Relationship between the characteristics of brain metastases and NSCLC types.

Metastatic lesion characteristics		NSCLC types, n (%)			p value
		Adenocarcinoma (n = 22)	Squamous cell carcinoma (n = 7)	Adenosquamous cell carcinoma (n = 4)	
No. of lesions	Soliter	7 (31.8)	5 (71.4)	1 (25.0)	0.872
	Multiple	15 (68.2)	2 (28.6)	3 (75.0)	
Rim enhanced	Yes	17 (77.3)	5 (71.4)	2 (50.0)	0.998
	No	5 (22.7)	2 (28.6)	2 (50.0)	
Vasogenic Edema	Yes	20 (90.9)	7 (100)	4 (100)	0.972
	No	2 (9.1)	0 (0)	0 (0)	
Hemorrhagic Lesion	Yes	2 (9.1)	1 (14.3)	1 (25.0)	1.000
	No	20 (90.9)	6 (85.7)	3 (75.0)	
Leptomeningeal Metastasis	Yes	0 (0)	0 (0)	0 (0)	N.A
	No	22 (100)	7 (100)	4 (100)	
Calcification	Yes	5 (22.7)	1 (14.3)	1 (25.0)	1.000
	No	17 (77.3)	6 (85.7)	3 (75.0)	

*Kolgomorov Smirnov test.

bleeding risk in squamous cell carcinoma but without consistent genotype correlation (13,14). Similarly, peritumoral vasogenic edema is a common feature of brain metastases across NSCLC subtypes and may reflect tumor brain microenvironment interactions

rather than EGFR status per se (15). Biologically, several mechanisms have been proposed to explain genotype-dependent differences in metastatic behavior. EGFR mutations have been implicated in epithelial-to-mesenchymal transition, altered cell

Table 5. Relationship between EGFR status and NSCLC types.

EGFR	NSCLC types, n (%)			p value
	Adenocarcinoma (n = 22)	Squamous cell carcinoma (n = 7)	Adenosquamous cell carcinoma (n = 4)	
Mutation type	7 (31.8)	4 (57.1)	3 (75.0)	0.509
Wild type	15 (68.2)	3 (42.9)	1 (25.0)	

*Kolgomorov Smirnov test.

Table 6. Relationship between EGFR mutation type and NSCLC types.

EGFR mutation types	NSCLC types, n (%)			p value
	Adenocarcinoma (n = 22)	Squamous cell carcinoma (n = 7)	Adenosquamous cell carcinoma (n = 4)	
Exon 19	6 (85.7)	2 (50.0)	3 (100)	0.995
Exon 21	1 (14.3)	2 (50.0)	0 (0)	

*Kolgomorov Smirnov test.

adhesion, increased migratory capacity and modulation of the blood–brain barrier, with supporting in vitro data showing differential downstream signaling (for example Akt and Erk) between Exon 19 deletions and L858R mutations (16,17). These molecular differences may underlie clinical observations that some EGFR mutation subtypes are associated with higher cumulative incidence or earlier onset of brain metastases, although published cumulative incidence estimates vary widely and are influenced by survival time, treatment exposure and detection practices (18–20). Our comparisons of imaging phenotypes by histologic subtype (adenocarcinoma, squamous cell carcinoma, adenosquamous carcinoma) did not reveal significant differences. This is consistent with studies reporting that adenocarcinoma more commonly gives rise to brain metastases but that, once present, imaging features such as peritumoral edema and the presence of hemorrhage may not be reliably distinct between subtypes on CT scan (21–24). The lack of leptomeningeal disease in our sample may reflect the modest sample size and the sensitivity limitations of CT scan for detecting subtle leptomeningeal involvement compared with MRI or CSF cytology (25). This study has several important limitations that affect interpretation and generalizability. First, the retrospective

single-center design and the relatively small final sample size limit statistical power and increase the risk of type II error; subgroup analyses by mutation exon and histologic subtype are particularly underpowered. Second, the use of MDCT rather than MRI for intracranial assessment may have reduced sensitivity for small lesions, leptomeningeal disease, and detailed characterization of peritumoral edema and enhancement patterns; MRI remains the preferred modality for neuro-oncologic imaging and may yield different findings. Third, selection bias is possible because only patients who underwent both head MDCT and EGFR testing (performed off-site in some instances) were included, and individuals without EGFR results or with non-evaluable imaging were excluded. This selective inclusion may have introduced systematic differences in the study population. Notably, the finding that most adenocarcinoma cases were EGFR wild-type which is atypical for an Asian cohort is likely explained by limited access to routine EGFR testing in our region. Molecular analysis requires shipping tissue samples to the capital city, resulting in substantial financial and logistical barriers. Consequently, EGFR status was obtained primarily in patients with longer survival and adequate financial resources, creating a highly selected, non-consecutive sample and

potentially skewing the mutation frequency distribution. Fourth, important clinical covariates that influence the risk and imaging characteristics of brain metastases including timing and type of systemic therapy, performance status, use of brain-directed treatment, and detailed smoking exposure information were unavailable or incompletely captured, limiting control of confounding factors. Finally, the small number of cases with specific EGFR mutation subtypes and less common histologic variants does not permit firm conclusions regarding subtype-specific imaging phenotypes. Given these limitations, the results should be interpreted cautiously. The findings were nevertheless consistent with the hypothesis that specific imaging characteristics of brain metastases, particularly rim enhancement, were more frequently observed in patients with wild-type lung tumors, although no overall association between the presence of brain metastasis and EGFR status in NSCLC patients was demonstrated. Future research should emphasize larger multicenter cohorts and prospective designs using MRI as the primary neuroimaging modality, coupled with complete molecular profiling and comprehensive treatment data. Such studies will be better positioned to determine whether reproducible imaging biomarkers of EGFR mutation status exist and whether they can meaningfully guide clinical decision-making or prognostication in NSCLC patients with brain metastases.

Conclusion

In this cohort of NSCLC patients with intracranial metastases assessed by head MDCT, rim enhancement was the only imaging characteristic significantly associated with EGFR status; EGFR-wild-type tumors were more frequently rim-enhancing, whereas EGFR-mutant tumors more often lacked rim enhancement. No significant relationships were identified between EGFR status and the number of lesions, vasogenic edema, hemorrhagic change, leptomeningeal disease or calcification. No significant associations were observed between NSCLC histologic subtype and the MDCT features analyzed, nor between histologic subtype and EGFR status in this sample.

Ethical Approval: The study protocol was reviewed and approved by the Biomedical Health Research Ethics Committee, Faculty of Medicine, Hasanuddin University, Makassar (No. 132/UN4.5.4.5.31/PP36/2025)

Conflict of Interest: Each author declares that he or she has no commercial associations (e.g., consultancies, stock ownership, equity interests, patent/licensing, arrangement etc-) that might pose a conflict of interest in connection with the submitted article.

Author contributions: DD — Conceptualization, Methodology, Investigation, Writing- Original Draft preparation; SA — Conceptualization, Methodology, Investigation, Resources, Validation, Writing- Original Draft preparation, Writing - Review & Editing, Supervision; JB — Validation, Writing - Review & Editing, Supervision; AAZ — Methodology, Formal Analysis; HI — Validation, Writing - Review & Editing; MI — Validation, Writing - Review & Editing; AY — Methodology, Investigation, Resources. All authors read and approved the final version of the manuscript.

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